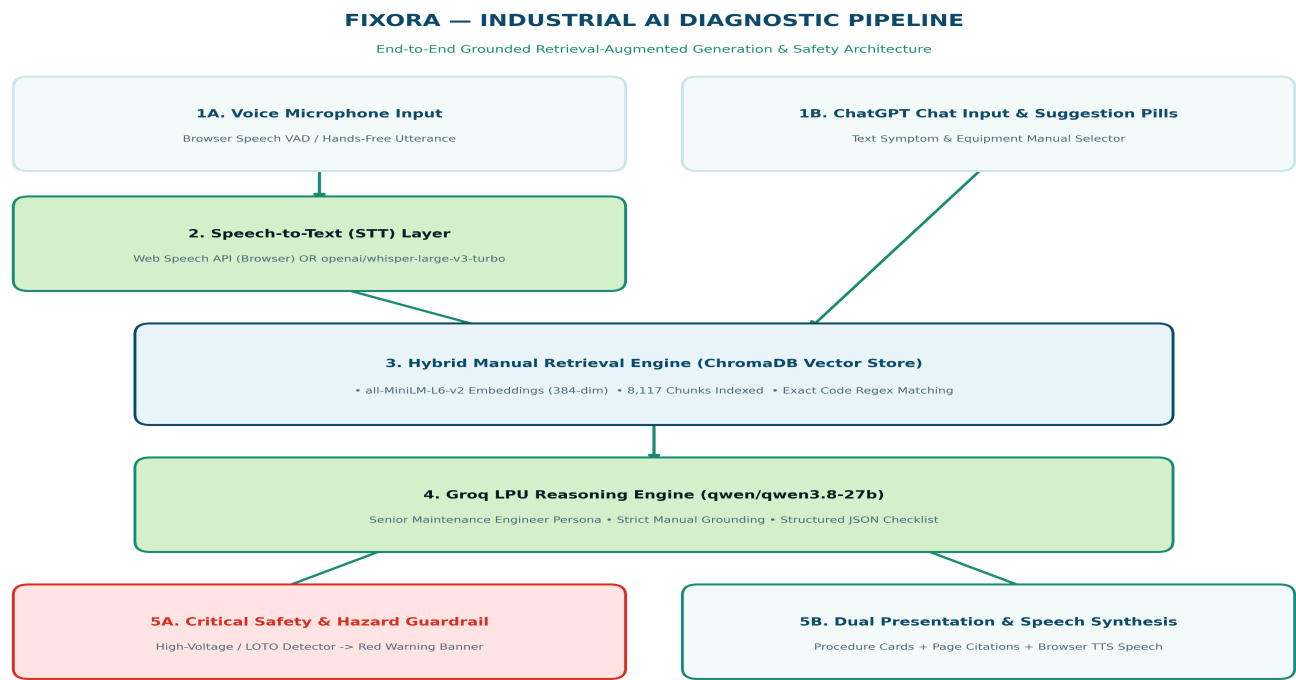


Fixora System Architecture & Pipeline Specification

Comprehensive Technical Reference · AI Models, Hybrid Retrieval & Safety Guardrails

Fixora is an industrial AI maintenance copilot engineered for biomedical and field service engineers. It converts multi-thousand-page technical service manuals into an instant, conversational, voice-enabled troubleshooting workspace. The architecture enforces deterministic grounding over **8,117 ChromaDB manual chunks** and leverages ultra-low latency Groq LPU inference.

1. High-Level Architectural Flowchart



2. Complete Inventory of AI Models & Engines

The table below details all models, runtimes, and roles across the 6 pipeline stages:

Component / Layer	Model Name	Provider / Runtime	Specifications & Function
Primary Reasoning LLM	qwen/qwen3.8-27b	Groq LPU Cloud	27B parameters. Senior engineer persona; generates structured JSON checklists in <1.5s.
Fallback LLM 1	qwen/qwen3.6-27b	Groq LPU Cloud	27B parameters. High-precision technical instruction following fallback.
Biomedical Fallback	allam-2-7b	Groq LPU Cloud	7B parameters. Multilingual medical and engineering diagnostic fallback.
Offline Edge Model	Qwen2.5-1.5B-Instruct	Local Ollama / CPU	1.5B parameters (4-bit GGUF) for 100% offline field laptop execution.
Semantic Embedder	all-MiniLM-L6-v2	sentence-transformers	384-dimensional dense semantic vectors optimized for technical text.
Vector Store Index	ChromaDB (v0.5+)	Local Embedded Store	8,117 dense chunks indexed with HNSW cosine similarity search.
Speech-to-Text (Cloud)	Web Speech API	Browser Native	Real-time speech capture with 1.3s Voice Activity Detection (VAD).
Speech-to-Text (Local)	whisper-large-v3-turbo	Hugging Face / PyTorch	High-fidelity offline transcription from raw uploaded audio files.
Text-to-Speech (Cloud)	SpeechSynthesis	Browser Native	Instant browser speech synthesis of conversational spoken summaries.
Text-to-Speech (Local)	facebook/mms-tts-eng	Meta MMS / PyTorch	Local high-quality neural voice synthesizer outputting WAV streams.
Web Fallback Agent	Tavily Search API	Tavily AI	Domain-restricted search over official manufacturer portals.

3. Safety & Grounding Guardrails

Fixora incorporates a dual-tier deterministic safety layer:

- **Hazard Interceptor:** Scans user queries and retrieved manual text for high-voltage, radiation, chemical, and pressure keywords (e.g. LOTO, lethal shock). When triggered, a high-priority red warning banner is injected at the top of the UI.
- **Anti-Hallucination Barrier:** The LLM is strictly constrained to output NOT_FOUND if the retrieved manual chunks do not contain verified technical evidence, preventing fabricated procedures.